



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

BT3303-1 – INDUSTRIAL BIOTECHNOLOGY
(For BT)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **How does modern biotechnology primarily differ from traditional biotechnology?**
A) Focuses only on fermentation techniques B) Utilizes genetic modification for industrial processes
C) Relies on natural selection for microbial improvement D) Uses only organic raw materials
2. **Which of the following is not a stage in the fermentation process?**
A) Inoculum development B) Downstream processing
C) Product purification D) Genetic sequencing
3. **Media formulation plays a crucial role in fermentation. Which of the following is an essential component?**
A) Nitrogen source B) Organic acids
C) Metal ions D) Alcohol
4. **Which technique is commonly used for the isolation of industrial microorganisms?**
A) Serial dilution B) Gram staining
C) Heat fixation D) Simple staining
5. **Why is cryopreservation widely used for microbial preservation?**
A) Eliminates all non-useful microbes B) It prevents contamination during fermentation
C) It allows long-term storage of microbial strains D) It enhances microbial growth rate
6. **Which method is used for microbial strain improvement?**
A) Random mutation B) Selective breeding
C) Genetic engineering D) All of the above
7. **Which factor significantly impacts fermentation economics?**
A) Cost of raw materials B) Availability of water
C) Atmospheric temperature D) Size of the industry
8. **The microorganism used in citric acid production is:**
A) *Saccharomyces cerevisiae* B) *Aspergillus niger*
C) *Lactobacillus acidophilus* D) *Acetobacter aceti*
9. **An enzyme commonly used in the detergent industry is:**
A) Amylase B) Lipase
C) Protease D) Cellulase
10. **What is the main function of amylase in industrial processes?**
A) Break down proteins B) Hydrolyze fats
C) Convert starch into sugar D) Improve fermentation efficiency

11. Which antibiotic is classified as a beta-lactam?
 A) Penicillin B) Streptomycin
 C) Erythromycin D) Rifampin
12. The microorganism commonly used for the production of erythromycin is:
 A) *Streptomyces erythraeus* B) *Bacillus subtilis*
 C) *Pseudomonas aeruginosa* D) *Aspergillus flavus*
13. A secondary metabolite among the following is:
 A) ATP B) Glucose
 C) Penicillin D) NADH
14. Nisin is classified as a:
 A) Biofertilizer B) Biopolymer
 C) Antibiotic D) Bio-preservative
15. What is the primary function of biopesticides?
 A) Increase plant growth B) Control pests naturally
 C) Reduce soil nutrients D) Improve crop coloration
16. An example of a biopolymer is:
 A) Xanthan gum B) Ethanol
 C) Citric acid D) Lysine
17. The recombinant therapeutic protein used for diabetes treatment is:
 A) Erythropoietin B) Humulin
 C) Growth hormone D) Peroxidase
18. Interferon- α is primarily used for treating:
 A) Diabetes B) Viral infections
 C) Bacterial infections D) Cancer
19. Which biotechnological input improves wool quality?
 A) Genetic modification of sheep B) Increasing dietary proteins
 C) Chemical treatment D) Selective breeding
20. What is the primary advantage of biotech-based silk production?
 A) Increased fiber strength B) Faster silk growth
 C) Reduced chemical use D) All of the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Classify the different types of fermentation processes and their stages in industrial applications.	8	L*2	1	2,4
	b) Analyze and explain the metabolic basis of product formation in industrial biotechnology by identifying key metabolic pathways and evaluating their role in industrial applications.	8	L4	1	2
2.	a) Outline the essential requirements for designing a fermentation medium.	8	L2	1	2
	b) Summarize the industrial-scale production process of Lactic acid and its applications.	8	L2	2	4
3.	a) Illustrate the industrial production of Lysine and its importance in biotechnology.	8	L2	2	4
	b) Demonstrate the microbial production of ethanol with a flowchart.	8	L2	2	
Unit – II					
4.	a) Describe the production process for amylase and its industrial significance.	8	L1	3	4
	b) Summarize the production process of Lipase and its role in the industry.	8	L2	3	4
5.	a) Outline the fermentation-based production of streptomycin antibiotic with a suitable flowchart.	8	L2	3	4

b)	Demonstrate the microbial process for the production of biopolymers like Xanthan gum.	8	L2	4	4
6. a)	Illustrate how single cell protein SCP can be compared to traditional protein sources in terms of efficiency and sustainability and add a note on its nutritional benefits.	8	L2	4	4
b)	Enlist the steps involved in the fermentation process of cheese production.	8	L1	4	4
Unit – III					
7. a)	What are the challenges and advancements in the production of recombinant therapeutic proteins?	8	L1	5	4
b)	Illustrate how the production process of Interferon- α is applied in clinical settings and its impact on patient treatment.	8	L2	5	4
8. a)	Summarize the economic and regulatory considerations in the production of modern biotechnology products.	8	L2	5	4
b)	Illustrate the microbial synthesis of Erythropoietin as therapeutic proteins.	8	L2	5	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
